Ankush Singh

Co-founder & CTO at Construct Computer | Full-Stack AI Agent Engineer | TypeScript, Cloudflare Workers, MCP

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Summary

Co-founder and CTO who builds AI agent products end to end, from the agent runtime and evals to the interface people use. Architected and wrote most of Construct Computer, where each AI employee gets its own cloud computer, on TypeScript, React, Cloudflare Workers, Durable Objects and the Model Context Protocol (MCP); it launched as **#1 Product of the Day on Product Hunt**. Shipping production software since 2022 across AI agents, LLM evals, developer tools and Solana, with a product acquired by Forward Research (BetterIDEa), 14 published npm packages (~19K downloads) and 13 hackathon and coding-competition wins.

Experience

Co-founder & CTO, Construct Computer | Remote, India | Feb 2026 - Present

A workforce of AI employees that each get their own cloud computer, on a desktop where you can watch them work and take over; paid plans from $9/month.

- Architected and wrote most of the platform: a TypeScript monorepo with an agent runtime on Cloudflare Workers and Durable Objects, a React desktop where users watch agents live over WebSockets, MCP apps, long-term memory and reusable workflows.
- Migrated the backend from a Docker-container-per-user VPS monolith to Cloudflare Workers, Durable Objects, Containers, R2 and D1 in ~10 days, retiring ~1,300 lines of container code so cost tracks work, not signups; later rebuilt it as v2 and moved every user.
- Hardened agent tool use: commands run in disposable sandboxes as a least-privilege user (no sudo, per-command elevation) with live user takeover; open-sourced Foxwatch, the Cloudflare monitor behind status.construct.computer.
- Shipped the Jev decision model into memory entity resolution 5 days after release: one 358 to 611 ms call replaced up to 3 calls to a JSON-schema-constrained Llama 4 Scout judge on Workers AI (now the fallback), at ~$0.017 per 1,000 decisions.
- First-authored Construct Technical Report 2026-01, a drop-in model-swap study of Clef and Clef-flash vs Jev (117 synthetic cases, 1,053 calls), and open-sourced CWAB, a deterministic benchmark running Construct, OpenClaw and Hermes Agent on 5 LLMs.
- Built and launched NepoNet (nepotism.network) within four weeks of the Construct launch: a warm-intro finder for X with a remote MCP server (OAuth), an OpenAPI tools API, llms.txt and an Agent Skill for coding agents; #9 Product of the Day.

  Tech: TypeScript, React, Hono, Cloudflare Workers, Durable Objects, Agents SDK, Sandboxes, Workers AI, Vectorize, D1, R2, MCP

Independent Developer, Self-employed | Remote, India | Nov 2025 - Feb 2026

- Built PumpMyClaw, a platform that runs autonomous AI trading agents (OpenClaw with 5 skills) on Solana and Monad, each in its own isolated Docker container, a design later carried into Construct's first version.
- Built pixels.earth, a real-time on-chain pixel canvas on Solana using MagicBlock ephemeral rollups and Anchor programs, and published create-magicblock, a CLI that scaffolds working MagicBlock apps (1,310 npm downloads).

  Tech: TypeScript, Rust, Anchor, Solana, MagicBlock, Docker, OpenClaw, React, Node.js

Entrepreneur in Residence, Forward Research (Arweave / AO) | Remote, India | Nov 2024 - Oct 2025

- Joined after Forward Research, the team behind AO, acquired BetterIDEa; rewrote the IDE for HyperBEAM with a rebuilt terminal and a GraphQL explorer (its @betteridea/codecell package has 3,717 npm downloads).
- Made the AO Package Manager (APM) part of AO's core: 6 merged PRs to permaweb/aos (2024 to 2025), including #454, which loads APM by default in every new aos process; the apm-tool CLI has 2,104 npm downloads.
- Built WAuth, a social-login SDK (Google, X, GitHub, Discord) with invisible non-custodial wallets (~5,300 npm downloads), later handed to ArLink Labs and rebuilt on Cloudflare Workers as ArlinkAuth.
- Shipped Subspace (Discord-style chat on AO), Visual AO (flow-based programming) and the ao-fetch and LuaX developer libraries.

  Tech: TypeScript, React, Next.js, Node.js, Lua, GraphQL, OAuth, Arweave, AO, HyperBEAM

Developer Relations Engineer, Arweave India | Remote, India | Jun 2024 - Apr 2025

- Organised Arweave Day India 2025, a multi-day summit and hackerhouse, and gave ADI Tour talks in Gurgaon and Chandigarh.
- Co-organised hackerhouse cohorts, built their cohort and agenda pages, mentored 100+ developers shipping on Arweave and AO, and judged 40+ builds at Longview Labs' Arweave Full Stack Hack.

  Tech: React, Next.js, Express.js, Tailwind CSS, Arweave, AO

Founder, BetterIDEa | Remote, India | Dec 2023 - Oct 2024

- Built BetterIDEa, a web IDE for Arweave and AO, at Arweave India's first hackerhouse, where it won the Grand Prize, and debuted it alongside the AO testnet launch (Feb 2024).
- Grew it into a developer tooling suite with npm packages, a CLI, documentation and the AO Package Manager (APM) through the three-month Arweave India Launchpad, and demoed it at Demo Day (Jul 2024); Forward Research then acquired it.

  Tech: TypeScript, React, Next.js, Lua, AO, Arweave

Software Engineer Intern, Physics Wallah | India | Nov 2022 - May 2023

- Built a live-class tool for faculty (New Products Initiative) in Electron, React and TypeScript with in-class questions, real-time YouTube chat parsing (10K+ messages) and analytics, used in classes with 50K+ concurrent viewers.
  Tech: Electron, React, TypeScript, Tauri, Python, Flask, YouTube Data API, AWS, Linux

Skills

AI & Agents: AI agents, Large language models (LLMs), Model Context Protocol (MCP), Tool calling (function calling), Agent memory & retrieval, Multi-agent orchestration, Sandboxed execution, Human-in-the-loop, Evals & benchmarks, LLM-as-judge, Structured outputs
Cloudflare: Cloudflare Workers, Durable Objects, Agents SDK, Workers AI, AI Gateway, Vectorize, D1, R2, KV, Queues, Sandboxes
Languages: TypeScript, JavaScript, Rust, Python, Lua, SQL
Full-Stack: Node.js, Hono, Express.js, REST APIs, OpenAPI, GraphQL, WebSockets, React, Next.js, Tailwind CSS, Electron
Web3: Solana, Anchor, Pinocchio, PDAs, SPL Token, MagicBlock, Arweave, AO, HyperBEAM, Solidity, EVM
ML & DevOps: ONNX Runtime Web, TensorFlow.js, WebGPU, YOLO, CLIP embeddings, OpenCV, Docker, GitHub Actions, CI/CD, AWS

Projects

TradeX (formerly PokePerps) | tradex.cards | Side project | Feb 2026 - Mar 2026

Non-custodial perpetual futures on Pokemon card prices on Solana mainnet, shut down in Mar 2026

- Wrote the USDC-margined Rust (Pinocchio) program with up to 20x leverage, a keeper oracle that published TCGplayer prices on-chain every 30 seconds, and an MCP server so AI agents could trade.
- Took the smart contract through an independent security audit and fixed all 12 findings (1 high, 5 medium, 6 low); the report is public at tradex.cards/audit-032026.pdf.
  Tech: Rust, Pinocchio, Solana, Bun, Elysia, SQLite, React, MCP

MyPokemonScanner | web.archive.org (mypokemonscanner.com) | Jan 2026

Pokemon card scanner that runs entirely in the browser, offline and without uploading images

- Trained a YOLO11n card detector (~10 ms) and matched MobileCLIP-S2 embeddings against 40K English and Japanese cards (50 to 80 ms on desktop), after testing CLIP, SigLIP and FP16/Int8 quantization.
  Tech: ONNX Runtime Web, TensorFlow.js, WebGPU, WASM, OpenCV, React, Vite

Open source | github.com/ankushKun

Packages, tools and upstream contributions

- Published 14 npm packages with ~19K downloads (top: @betteridea/codecell, @wauth/sdk, apm-tool, @construct-computer/app-sdk) and merged PRs to permaweb/aos (6), ao-cookbook, ar-io/docs, DefiLlama-Adapters and Wander.
  Tech: TypeScript, Lua, Cloudflare Workers, Node.js

Awards

- **#1 Product of the Day and #5 Product of the Week in Productivity**, Product Hunt, Construct Computer launch, Aug 2026
- **#9 Product of the Day**, Product Hunt, NepoNet (Nepotism Network) launch, Sep 2026
- **Selected for the Build Sprint**, TinyFish Accelerator, for founders building AI agents (Construct Computer), Mar 2026
- **Coinbase Developer Platform 3rd place and 2 pool prizes**, ETHIndia 2024, Dec 2024
- **3rd prize**, Weavers Raid III, for DumDumDeploy (later Arlink), Aug 2024
- **Grand Prize**, Arweave India Hackerhouse Cohort 1, Bangalore, for BetterIDEa, Dec 2023
- **Tableland pool prize**, ETHOnline 2023 (ETHGlobal), for GitRaven, Oct 2023
- **2nd place**, Hack-O-Octo 1.0, for GitRaven, Oct 2023
- **1st prize**, C-DAC Vegathon finals at IIT Guwahati, flight controller on the VEGA processor, Jun 2023
- **1st place**, IIT Roorkee Intel oneAPI hackathon, Mar 2023

Publications & Talks

- **Construct Technical Report 2026-01, "Compatible APIs, Incompatible Thresholds"**: First author, with Nischal Naik. Clef and Clef-flash vs Jev on five production decision specs, 1,053 calls. construct.computer/blog/clef-vs-jev-benchmark, Oct 2026
- **Construct blog**: 11 posts on agent infrastructure, evals and the product, including running agents on Cloudflare and Jev in memory. construct.computer/blog
- **Publish to npm from GitHub Actions using OIDC**: Tutorial on tokenless trusted publishing. ankush.one/blogs/npm-oidc-publishing
- **Talks and video**: ADI Tour developer talks (2025), product demos and tutorials. youtube.com/@AnkushKun

Education

Bachelor of Engineering (B.E.) in Computer Science, Chandigarh University, Punjab, India